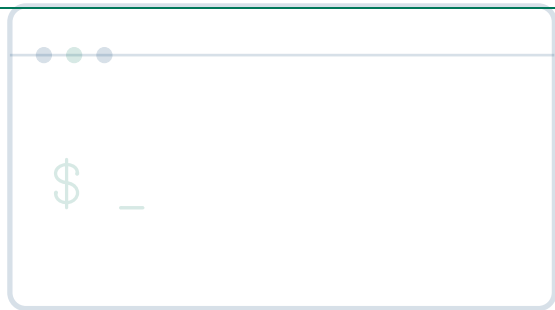


Linux for Computational Chemistry

Lecture 1: Introduction to Linux and the Command Line

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Why Linux?

Why does computational chemistry run on Linux?

- Virtually every HPC cluster and supercomputer runs Linux
- Quantum chemistry and MD packages (Gaussian, ORCA, Gromacs, VASP, AutoMeKin. . .) are built, tuned, and often only distributed for Linux
- The shell lets you **automate** repetitive tasks: hundreds of single-point calculations, conformer scans, batch post-processing
- Reproducibility: a script is a precise, shareable record of what you did
- Skills transfer directly to any cluster you will ever log into

What is “Linux” exactly?

- **Kernel**: the core program managing hardware, processes, memory
- **Distribution** (Ubuntu, CentOS/Rocky, Debian...): kernel + tools + package manager
- **Shell** (bash, zsh): the program that reads and executes your typed commands
- You almost never touch the kernel directly — you talk to the **shell**

Note

Most HPC clusters use CentOS/Rocky Linux with the `bash` shell. Everything in this course targets `bash`.

Anatomy of the prompt

```
[jperez@login1 ~]$
```

- jperez — your username
- login1 — the hostname (the login node of a cluster)
- ~ — current directory (here, your home)
- \$ — ready for input (a # would mean root!)

The Filesystem

The filesystem is a single tree

- Everything starts at the root, / — there is no “C:\
• Key locations you will meet on a cluster:

<code>/home/you</code>	your personal, backed-up, quota-limited space
<code>/scratch</code> or <code>/work</code>	large, fast, <i>not</i> backed-up space for jobs
<code>/usr</code> , <code>/opt</code>	installed software
<code>/tmp</code>	temporary files, cleared periodically
<code>/etc</code>	system configuration

Tip

Always run large quantum chemistry jobs from `/scratch`, never from `/home` — it is faster and avoids filling your quota.

Absolute vs. relative paths

- **Absolute path:** starts from /, always correct regardless of where you are
- **Relative path:** interpreted from your **current** directory

```
/home/jperez/project/butanol/opt.log # absolute  
project/butanol/opt.log # relative  
. # this directory  
.. # the parent directory  
~ # your home directory
```

Moving around: pwd, ls, cd

```
pwd # print working directory
ls # list contents
ls -l # long format: permissions, size, date
ls -a # include hidden files (dotfiles)
ls -lh # long format, human-readable sizes
ls -lt # sort by modification time
cd project/butanol # move into a subdirectory
cd .. # move up one level
cd # go home
cd - # go to the previous directory
```

Exercise 1.1

Starting from your home directory:

1. Print your current working directory.
2. Create the path `project/butanol/opt1` in one command (hint: look up `mkdir -p`).
3. Move into `project/butanol/opt1` using a relative path.
4. From there, return to your home directory in *two different ways*.
5. List the contents of `project`, including hidden files, sorted by modification time.

Solution

```
pwd
mkdir -p project/butanol/opt1
cd project/butanol/opt1
cd ~ # or: cd
cd $HOME # or: cd - (after having moved once)
ls -lat project
```

Files and Directories

Creating, copying, moving, removing

```
mkdir results # create a directory
rmdir results # remove it, only if empty
touch notes.txt # create an empty file / update timestamp
cp opt.log opt.bak # copy a file
cp -r butanol butanol_bak # copy a directory recursively
mv opt.bak archive/ # move / rename
rm notes.txt # delete a file (no recycle bin!)
rm -r butanol_bak # delete a directory recursively
```

Note

`rm` is permanent. There is no undo. Double-check the target, especially with `-r` and wildcards.

Wildcards

```
*.log # any name ending in .log  
opt?.log # opt1.log, opt2.log... (exactly one character)  
conf[0-9].xyz # conf0.xyz ... conf9.xyz  
conf{1,2,5}.xyz # exactly conf1.xyz, conf2.xyz, conf5.xyz
```

Tip

Before running `rm *.log`, run `ls *.log` first to see exactly what will be deleted.

Exercise 1.2

1. In `project/butanol/opt1`, create three empty files: `conf1.xyz`, `conf2.xyz`, `conf3.xyz`.
2. Copy all three into a new subdirectory `backup` in a single command using a wildcard.
3. Rename `conf2.xyz` to `conf2_final.xyz`.
4. Remove the `backup` directory and everything in it.

Solution

```
cd project/butanol/opt1
touch conf1.xyz conf2.xyz conf3.xyz
mkdir backup
cp conf*.xyz backup/
mv conf2.xyz conf2_final.xyz
rm -r backup
```

Looking inside files

```
cat opt.log # dump entire file to screen
less opt.log # scroll through a file (q to quit, / to search)
head opt.log # first 10 lines
head -n 20 opt.log # first 20 lines
tail opt.log # last 10 lines
tail -f slurm.out # "follow" a growing file live (Ctrl+C to stop)
wc -l opt.log # count lines
```

Tip

`tail -f` on a running job's output file is the standard way to watch a calculation progress in real time.

Getting Help

Getting help without leaving the terminal

```
man ls # full manual page  
ls --help # short usage summary  
info coreutils # more detailed manuals (less common)  
apropos partition # search manual page descriptions
```

Note

When in doubt about a command's options, `man` is always the first place to check — faster and more reliable than searching online, and it works with no internet on a compute node.

Permissions

Why permissions exist

- Linux is a **multi-user** system — on an HPC cluster, hundreds of people share the same machine
- Every file and directory has an **owner** (a user) and a **group**, plus rules for everyone else
- Permissions decide who can read your inputs, overwrite your results, or run your scripts
- Get them wrong and either: your labmates cannot read shared results, or a script silently refuses to run

The three classes and three actions

Every file/directory defines permissions for **three classes** of people:

owner (u)	the user who created the file
group (g)	members of the file's group
others (o)	everybody else on the system

... and each class can independently have:

r (read)	view contents / list a directory
w (write)	modify contents / add-remove files in a directory
x (execute)	run as a program / <i>enter</i> a directory (cd into it)

Reading 1s -1, character by character

```
-rwxr-xr-- 1 jperez chemgroup 4096 Jun 3 10:21 run_opt.sh
```

1	2-4	5-7	8-10
-	r w x	r - x	r - -
type	owner	group	others

- Character 1: -=regular file, d=directory, l=symbolic link
- Then *three blocks of three*: owner, group, others — always in the fixed order r, w, x
- A - in a slot means that permission is *absent*, not “zero users”
- So here: the owner jperez can read/write/execute; the group chemgroup can read/execute; everyone else can only read

r, w, x mean something different for directories!

	on a file	on a directory
r	view its contents (cat)	list its contents (ls)
w	modify/overwrite it	create/delete/rename entries inside
x	run it as a program	cd into it, access files inside by name

Note

A directory with r but no x is a common trap: you can list filenames with ls, but you cannot cd into it or open any file inside it by path. You almost always want r and x together on directories.

Octal notation: the same thing as numbers

Each permission has a numeric weight, and the three weights for a class simply **add up**:

$$r = 4 \quad w = 2 \quad x = 1$$

digit	sum	symbolic	meaning
7	4+2+1	rwx	read, write, execute
6	4+2	rw-	read, write
5	4+1	r-x	read, execute
4	4	r-	read only
0	-	-	nothing

One digit per class, always in the order **owner-group-others**: `chmod 750` means `rwX`
`r-X --`.

Changing permissions: chmod

```
chmod +x run_opt.sh # symbolic: add execute for everyone
chmod u+x,g-w run_opt.sh # owner +execute, group -write
chmod go-rwx secret.dat # remove ALL access for group and others
chmod 750 run_opt.sh # octal: rwx r-x ---
chmod 644 opt.log # octal: rw- r-- r--
chmod 700 private_notes.sh # octal: only the owner can do anything
```

Tip

Symbolic (u/g/o + +/- + r/w/x) is good for *tweaking* one class. Octal is good for *stating the full picture* in one number — most people default to octal once they are comfortable with the r=4, w=2, x=1 table.

Ownership: `chown` and `chgrp`

```
ls -l opt.log # see current owner:group  
chown jperez opt.log # change the owner (often needs sudo)  
chgrp chemgroup opt.log # change the group  
chown jperez:chemgroup opt.log # change both at once
```

Note

On a shared cluster you will rarely run `chown` yourself — files you create are automatically owned by you and your default group. It is still useful to *recognize* the command when you see it in documentation or scripts.

Exercise 1.3 — Permissions

Exercise 1.3

1. Create a file `run_opt.sh` and inspect its default permissions with `ls -l`. Identify owner, group, and others' permissions separately.
2. Make it executable for yourself only (not group, not others), using octal notation. Predict the 10-character string `ls -l` will show *before* you run it.
3. Verify the change with `ls -l`.
4. Now make it readable and executable by everyone, but writable only by you, using symbolic notation.
5. Create a directory `private` with permissions 700, put a file inside it, then remove *only* the execute permission for yourself (`u-x`) and try `cd private` and `ls private`. Which one fails, and why?

Solution 1.3

Solution

```
touch run_opt.sh
ls -l run_opt.sh # -rw-r--r-- (typical default)
chmod 700 run_opt.sh # predict: -rwx-----
ls -l run_opt.sh
chmod go+rx,go-w run_opt.sh # equivalently: chmod 755 run_opt.sh

mkdir -m 700 private
touch private/note.txt
chmod u-x private
cd private # fails: Permission denied (no x = cannot enter)
ls private # still works: r lets you list names inside
```

Searching for Files

find: locating files anywhere in a tree

```
find . -name "*.log" # search recursively by exact name pattern
find . -iname "*.LOG" # case-insensitive
find . -maxdepth 2 -name "conf*.xyz" # limit how deep it recurses
find . -type d -name "opt*" # only directories
find . -type f -mtime -1 # regular files modified in the last day
find . -size +100M # files larger than 100 MB
```

Note

Unlike `ls`, `find` searches an entire directory tree at once and can filter by type, age, or size — essential once a project has hundreds of subdirectories.

find with -exec: act on what you find

```
find . -name "*.chk" -exec rm {} \;  
find . -name "*.log" -exec grep -l "Error termination" {} \;  
find . -name "*.log" -mtime +7 -delete
```

- {} is replaced by each matching path; the command ends in \;
- -delete is a shortcut for the common case of removing what matched

Tip

Always run the find command *without* -exec rm or -delete first, to see exactly what would be selected.

Exercise 1.4

Inside your `2-butanol_project` tree (create a few dummy `.log` and `.chk` files to test with):

1. Find every `.log` file anywhere under the project, regardless of depth.
2. Among those, find only the ones that contain the text "Error termination" (combine `find` with `-exec grep -l`).
3. List every file larger than 10 MB anywhere in the tree.
4. Find, but do *not yet delete*, every `.chk` file older than 7 days. Only once you have checked the list, re-run with `-delete`.

Solution

```
find . -name "*.log"
find . -name "*.log" -exec grep -l "Error termination" {} \;
find . -type f -size +10M
find . -name "*.chk" -mtime +7 # inspect first
find . -name "*.chk" -mtime +7 -delete # then delete
```

Process and Job Control

Foreground, background, and interrupting

```
g16 < opt.com > opt.log & # launch in the BACKGROUND (note the &)  
jobs # list background jobs of this shell  
fg # bring the most recent job to the foreground  
fg %1 # bring job number 1 to the foreground  
bg %1 # resume a suspended job in the background
```

- Ctrl+C sends an interrupt — normally terminates the foreground process
- Ctrl+Z *suspends* the foreground process (pauses it, does not kill it)

Inspecting processes: ps and top

```
ps # your processes attached to this terminal
ps aux # every process on the system, full detail
ps aux | grep g16 # is my Gaussian job still running?
top # live, auto-updating view (press q to quit)
```

Note

On an HPC *login* node, ps/top show only what is running there — never long calculations, which belong on compute nodes via the scheduler (Lecture 4).

Stopping a process: kill

```
kill 12345 # ask process 12345 (its PID) to terminate gracefully  
kill -9 12345 # force-kill, when it refuses to stop  
killall g16 # kill by program name instead of PID
```

Tip

Always try a plain `kill` first (it lets the program clean up, e.g. close files properly). Reach for `kill -9` only if the process is genuinely stuck.

Surviving logout: nohup

```
nohup ./run_opt.sh > run.log 2>&1 &
```

Note

Without `nohup`, a background job launched on a login node is killed the moment you disconnect. In practice, real quantum chemistry jobs are submitted through SLURM (Lecture 4) precisely so you never have to rely on keeping a terminal open — but `nohup` is essential to know for quick, short tasks on a login node.

Exercise 1.5

1. Launch `sleep 300` (a command that just waits 5 minutes) in the background.
2. Confirm it is running with `jobs` and with `ps`.
3. Bring it to the foreground with `fg`, then suspend it with `Ctrl+Z`.
4. Resume it in the background with `bg`, and confirm with `jobs` that it is running again.
5. Finally, terminate it early with `kill` instead of waiting 5 minutes.

Solution

```
sleep 300 &  
jobs  
ps | grep sleep  
fg # Ctrl+Z to suspend once in the foreground  
bg  
jobs  
kill %1 # or: kill <PID> from ps
```

Text Editors: nano and vi/vim

Surviving with nano

```
nano input.com
```

- Just type to edit — no modes to worry about
- Ctrl+O save (“WriteOut”), Ctrl+X exit, Ctrl+K cut a line
- Simple and safe, but limited — vi/vim is worth learning next

Why learn vi/vim?

- `vi` is installed on *every* Unix/Linux system without exception — on a bare cluster node, it may be the only editor available
- `vim` (“Vi IMproved”) is the version found on almost all modern distributions and clusters, and is what we will use here
- It is **modal**: keys do different things depending on which mode you are in — this is the single biggest adjustment for newcomers

The two modes that matter today

Normal mode keys are *commands* (move, delete, save, quit...)

Insert mode keys are typed as *text*, like a normal editor

- vim always opens in **Normal** mode
- Press `i` to enter **Insert** mode and start typing
- Press `Esc` to go back to **Normal** mode — you must do this before any command (including saving or quitting)

Note

The most common beginner mistake: typing `:wq` while still in Insert mode, which just inserts the literal characters `:wq` into the file. Always press `Esc` first.

Opening, saving, quitting

```
vim input.com
```

- i enter Insert mode at the cursor
- Esc return to Normal mode
- :w write (save) the file
- :q quit (fails if there are unsaved changes)
- :wq write and quit
- :q! quit *without* saving, discarding all changes

Tip

All `:-commands` are typed in Normal mode and confirmed with `Enter`. If you ever feel stuck, press `Esc` then type `:q!` and `Enter` to bail out safely.

Moving around and basic editing in Normal mode

<code>h j k l</code>	left, down, up, right (arrow keys also work)
<code>0 / \$</code>	start / end of the current line
<code>gg / G</code>	first / last line of the file
<code>x</code>	delete the character under the cursor
<code>dd</code>	delete (cut) the whole current line
<code>yy</code>	yank (copy) the whole current line
<code>p</code>	paste after the cursor
<code>u</code>	undo (repeat for more)
<code>Ctrl+r</code>	redo

Search and replace in vim

```
/SCF Done # search forward for "SCF Done", Enter to jump to it  
n # jump to the next match  
N # jump to the previous match  
:%s/old/new/g # replace ALL occurrences in the whole file  
:s/old/new/ # replace first occurrence on the current line
```

Note

This is the exact same substitution syntax as sed from Lecture 2 — vim and sed share the same regex heritage, so what you learn in one transfers to the other.

Exercise 1.6 — A first vim session

Exercise 1.6

1. Open a new file `notes.txt` with `vim`. Confirm you start in Normal mode.
2. Enter Insert mode and type three lines of arbitrary text. Return to Normal mode.
3. Move to the first line with `gg`, delete it with `dd`.
4. Undo that deletion with `u`.
5. Search for a word that appears in your text with `/`, then replace every occurrence in the file with `:%s/.../.../g`.
6. Save and quit with `:wq`. Re-open the file with `cat` to confirm your changes were saved.
7. Finally, open it again in `vim`, make an edit you do *not* want to keep, and exit discarding it with `:q!`.

Solution

```
vim notes.txt
# i, type three lines, Esc
gg
dd
u
/someword (Enter)
:%s/someword/otherword/g
:wq
cat notes.txt
vim notes.txt
# make an edit, then:
:q!
```

- You can navigate, inspect, create, copy, move, and remove files and directories
- You understand absolute vs. relative paths and the standard HPC directory layout
- You can read and set permissions
- You know how to ask the system itself for help

Homework — Build a project tree

Exercise 1.4 (Homework)

Build the following directory structure for a (fictitious) computational study of 2-butanol radical reactions, using a single sequence of shell commands:

```
2-butanol_project/  
|-- inputs/  
|-- outputs/  
|-- scripts/  
'-- README.txt
```

Then:

1. Write one line of text into `README.txt` describing the project (use `nano`, or research the `echo "..."` > file redirection we will cover next class).
2. Make everything in `scripts/` eventually executable-ready: set directory permissions to 755.